



From regulatory text to *operational action.*

An agentic compliance engine that turns a SEBI master circular into a living, auditable obligation register for one specific market intermediary — demonstrated on a real listed stock broker, with a pipeline that executes in production.

Regulation is text for humans. Compliance systems need *machine-checkable rules*.



PAIN 1 · INTERPRETATION

When SEBI issues or amends a circular, a human reads the PDF, decides whether it binds their firm, maps it to internal processes, and updates them. The work is slow, manual, and **divergent** — two firms read the same clause differently, and the regulator inherits the inconsistency.



PAIN 2 · PROOF

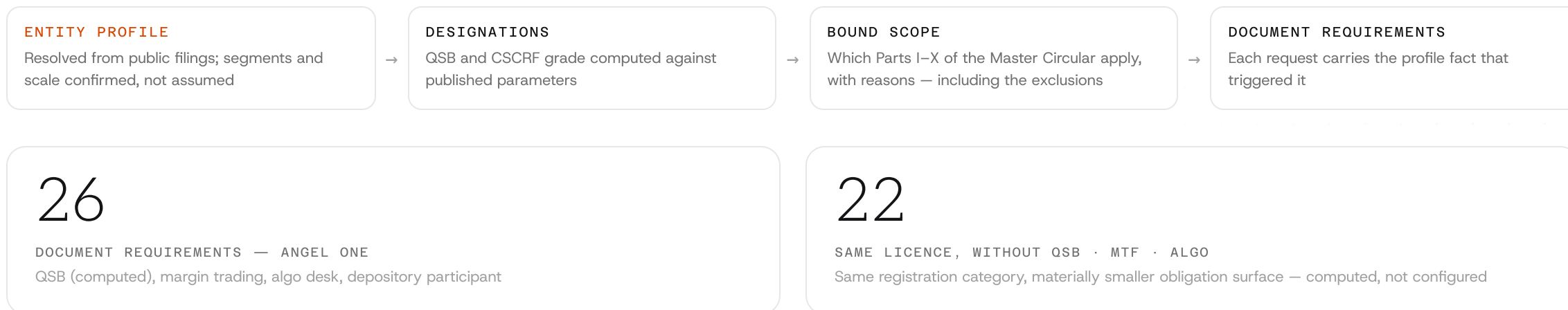
Compliance teams must track every standing obligation, hold **evidence** that each is fulfilled, and produce audit trails on inspection. Smaller intermediaries carry the same rulebook with the least capacity — the firms with the thinnest compliance teams hold the most gaps.



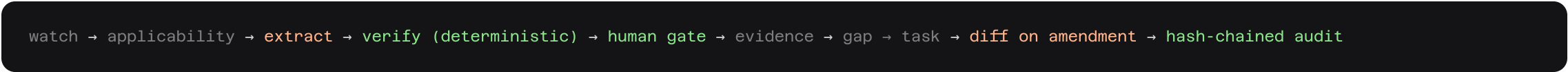
The bridging problem — unstructured regulatory text → structured, auditable, operational compliance — is what PS2 names as the core unsolved problem. The suggested corpus, the **Master Circular for Stock Brokers**, is the corpus this prototype runs on.

Two brokers holding the same licence *do not owe the same things*.

Most tooling stops at circular → obligation table: the same table for every firm. But a firm's real obligations are a function of what it is — its business segments, its scale, its designations. Poneglyph starts from the firm, not from a checklist: the entity profile determines which Parts of the rulebook bind, and the bound scope generates the document requirements.



A new regulation is a *new job* through a fixed pipeline.



WATCH / DECIDE	A scraper polls SEBI daily; an applicability agent rules whether each instrument binds this firm — quoting the clause that decides, including for the No .	HUMAN GATE	Drafts hold at pending-review. A named compliance officer approves or rejects; nothing reaches the register any other way — enforced structurally, the drafting path cannot write to it.
EXTRACT	Each obligation is drafted grounded to its source: circular, paragraph, and the exact character span of the clause.	AMEND / DIFF	On amendment the engine diffs old against new and re-maps only what changed; untouched obligations are never re-reviewed.
VERIFY	Five deterministic checks in plain code — citations resolve verbatim, deadlines parse, applicability matches, schema validates, chain appends. Any failure stops the run.	AUDIT	Every action appends to a SHA-256 hash chain. Integrity is verified by recomputation from content, so alteration of any record is detected by arithmetic.

Agents are ephemeral workers inside the pipeline; the typed obligation ontology is the permanent artefact. This is the only agent architecture a regulator can audit: every step recorded, every judgement replayable, every write signed.

Angel One Limited, onboarded from *its own public filings*.

₹6,201.98 cr

NET WORTH · FY2025-26

Total equity, standalone — XBRL filing (prior year ₹5,597.87 cr)

₹5,054.07 cr

REVENUE · FY2025-26

Standalone — same XBRL filing

3.86 cr

CLIENT BASE · JUN 2026

38.59 mn, business update — up 18.8% YoY

≈6.76 mn

NSE ACTIVE CLIENTS

Derived: 14.79% share × 4.57 cr market actives — an estimate, labelled as one

REAL · PUBLIC AND CHECKABLE

Identity (ISIN INE732I01021 · NSE ANGELONE · BSE 543235), listing, net worth, revenue, client base, market share — every fact carries the filing or business update it came from. Anything the engine computed itself is labelled **derived**.

ILLUSTRATIVE · SIMULATED POSTURE

All compliance posture — obligations met or gapped, evidence, tasks, audit events — is simulated and asserts nothing about Angel One's actual compliance. No registration number is invented: the public INZ000161534 is held as **declared** until its certificate is supplied. QSB is **computed against the published parameters and held unconfirmed** — the engine raises a document request for the exchange's designation letter instead of assuming.

43 obligations across 9 of 10 Parts of the Master Circular.

PART	TITLE — VERBATIM FROM THE CIRCULAR	OBLIGATIONS
I	Registration of Stock Brokers	3
II	Supervision & Oversight — system audit, QSB	2
III	Dealings with Client — the largest Part	19
IV	Technology Related Provisions — CSCRf, algo, DMA	6
V	Change in Status, Constitution, Control, Affiliation	2
VI	Foreign Accounts Tax Compliance Act Related Provisions	1
VII	Investor Grievance Redressal — SCORES, ODR	3
VIII	Default Related Provisions	0 — by determination
IX	Miscellaneous — outsourcing, books, upstreaming	5
X	Reporting Requirements	2

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GROUNDING INVARIANT

excerpt === paraText.slice(charStart, charEnd)

Holds for all 43 register entries — verified mechanically, not by eye. Any control walks back to the exact characters of the clause it descends from.

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THE ZERO IS A DECISION

Part VIII binds only upon a default event. The engine excluded it with the reason on the record and holds it under a standing trigger watch — an explained zero, not a gap.

Taxonomy read from SEBI/HO/MIRSD/MIRSD-PoD-1/P/CIR/2024/53 — the source circular, not a summary. Overlaid: CSCRf (graded by entity size) and QSB (enhanced obligations).

SEBI amended Para 46. Ten obligations re-mapped, *thirty-three untouched*.

Jul 3, 2026

CUSPA CIRCULAR ISSUED

HQ/38/11/(9)2026-MIRSD-POD/I/15382/2026 — pledge-based regime for clients' unpaid securities

10 / 33

RE-MAPPED / UNTOUCHED

The diff re-maps only what changed; nobody re-reviews the whole rulebook

Jan 2027

PHASED DEADLINES TRACKED

Every due date taken from the circular's own effectivity phases

The Watchtower caught the circular, the applicability agent ruled it binding with the clause quoted, the diff isolated the changed paragraphs, extraction drafted ten new obligations — pledge account, client intimation by SMS and e-mail, five-trading-day window with day-six release, daily reconciliation — and the verifier passed them to the human gate. Remediation tasks carry owners and the circular's own dates. **This is Pain 1, closed end to end, on a real nine-day-old amendment.**

Contrast case shown in-product: a mutual-fund TER circular is ruled **not applicable** — addressed to AMCs, and this firm holds no AMC registration. The engine files the reason it ignored an instrument; the No is audited like the Yes.

The pipeline *executes*. These results are observed, not claimed.

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CHAIN INTEGRITY — RECOMPUTED

Every audit event is SHA-256 hashed over its content plus the previous hash. Verification **recomputes from content**: the in-product tamper test edits one stored record, and verification then fails at that exact index — stored `c275c5cefec9` $\neq$ recomputed `898698aa53d8`. Independently re-derived outside the product: 61 events, 0 mismatches.

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VERIFIER — REJECTS FABRICATION

Citations resolve by exact string match against the submitted clause; character offsets are derived from the match. A fabricated excerpt fails and the run stops. So does a **one-word paraphrase** — which is how hallucination actually presents. Nothing failing a check can reach the register.

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AUTHORISATION GATE — PERSISTED

Approve and Reject mutate state in the session store. An approved obligation enters the register as **gap** — approved, no evidence bound yet — never as met. A second decision on the same draft returns 409. The drafting path has no write access to the register.

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EXTRACTION — LIVE MODEL CALL

One serialised LLM call per run, grounded to computed offsets. Observed latency 11s–500s+ on free-tier capacity; a run orphaned by the edge runtime is settled at read time as an **honest failure** rather than left spinning. No fabricated results on timeout, ever.

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Per-visitor sandboxes (session-keyed store), seeded from the same data — one reviewer's approvals never leak into another's view. The seeded register, evidence artifacts and tasks elsewhere in the sandbox remain simulated and labelled as such.

One obligation graph, *two windows*.

BROKER CONSOLE

The compliance team's working surface: register, evidence vault, document vault, remediation queue, pipeline runs. Gaps become tasks with owners and the circular's own deadlines. The officer stops reading PDFs and clears a queue.

INSPECTOR VIEW · READ-ONLY

The same data under a regulator's lens — enforced read-only by the interface. Any control traces to its clause, its evidence, and its tamper-evident history in three clicks; what once needed a data room and a week of correspondence becomes navigation.

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MCP SURFACE — MACHINE-QUERYABLE COMPLIANCE

The register is also a protocol: seven read-only tools over the ontology, with an inspector-scoped token in place of a data-room export. A regulator's own agent asks *"is this firm compliant with the CUSPA circular?"* and receives an answer traceable to clauses — and the inspector's queries are themselves appended to the audit chain. Because the ontology is open, every firm publishing it becomes comparable: the same question returns comparable answers, which is PS2's divergent-interpretation pain addressed at the root.

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executes in a Worker.

```
└─ Cloudflare edge ───────────────────────────────────────────────────┐
  /api/* → Worker    pipeline · verifier · gate · SHA-256 chain · KV sessions
  /*      → static assets  14 routes, prerendered — an API fault cannot take the site down
└────────────────────────────────────────────────────────────────────────┘

extraction → OpenAI-compatible endpoint (Kimi K3) · serialised · honest timeout
```

FRONTEND	Next.js 16 static export — no server runtime, no cold starts	ONTOLOGY	Typed obligation schema — the permanent artefact; open by design	SUPPLY CHAIN	0 npm audit findings; secrets held as Worker bindings, never in the bundle
API	Cloudflare Worker, Web Crypto, KV-backed per-visitor sessions	DELIVERY	Deploys on push to main via Cloudflare Workers Builds; one-command manual path	PROVIDER-AGNOSTIC	Extraction speaks the OpenAI wire format — the model is a config value

A forced-purchase market, *under-served at the small end.*

WHO PAYS

India's thousands of small and mid-size stock brokers, RIAs and portfolio managers — bound by the same rulebook as the largest firms, priced out of Big-4 compliance. Compliance spend is non-discretionary and deadline-driven.

MODEL — OPEN CORE

The obligation ontology is open: it sets the standard and makes compliance comparable across firms. The engine, evidence connectors and audit vault are the paid per-intermediary subscription, priced an order of magnitude below manual compliance.

EXPANSION

The engine is corpus-driven: the same pipeline points at the LODR, RA/IA and PMS rulebooks with no structural change — then at other Indian regulators. Each new corpus is an ingestion job, not a rebuild.

+ Alignment with the mandate: stronger compliance at thousands of small intermediaries is investor protection delivered through infrastructure. A shared open ontology also gives the regulator something it cannot buy today — **the same question, answered comparably, across every firm that publishes it.**

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Deployed, verified, *open to inspection.*

14

ROUTES LIVE

Entry gate, onboarding, register, document vault, live pipeline, inspector view

43

OBLIGATIONS ON REGISTER

9 of 10 Parts; every entry clause-grounded

56

SEEDED AUDIT EVENTS

SHA-256 chained; live events append per session

28

DOCUMENT REQUIREMENTS

Each carrying the profile fact that triggered it

LIVE PROTOTYPE	poneglyph.walrussecuritas.com	NEXT	Full-corpus ingestion at scale · evidence connectors to exchange and depository systems of record · pilot with a mid-size broker
LIVE PIPELINE	poneglyph.walrussecuritas.com/live		
REPOSITORY	github.com/walrus-securitas/poneglyph	TEAM	Anshuman Atrey — engineering & security · Gayatri Jaiswal — operations

Agents propose. A deterministic verifier checks. A named human signs. A regulator can inspect all of it — that is **agentic compliance built the way a regulator can trust.**